

15.439 Quantitative Investment Management

Assignment 6

April 24, 2026

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The key issue is that “guaranteed loser” can mean different things. Some strategies lose at the fixed nine-month horizon in every state of the world; others only become loss-making after repetition or under an asymptotic regime. As the assignment asks for a strategy guaranteed to lose money, it is useful to state clearly which notion is being used in each example.

1 Preliminaries

1.1 Formalized problem

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a finite horizon T of nine months. Any admissible strategy induces a terminal profit-and-loss random variable

$$\Pi_T : \Omega \rightarrow \mathbb{R}.$$

We will distinguish between one-shot loss at horizon T and asymptotic loss along a sequence of strategies $(\Pi_T^{(n)})_{n \geq 1}$. For this assignment, the strongest and most relevant notion is loss at the fixed horizon itself; the asymptotic notions are included only to classify examples that are often described as “guaranteed losers” even though the guarantee only appears in the limit.

1.2 Notions of loss

Definition 1 (Almost-sure loss at the horizon). *A strategy is an almost-sure loser if*

$$\mathbb{P}(\Pi_T < 0) = 1.$$

Definition 2 (Almost-sure convergence to loss). *A sequence $(\Pi_T^{(n)})_{n \geq 1}$ converges almost surely to loss if there exists a random variable $\Pi_T^{(\infty)}$ such that*

$$\Pi_T^{(n)} \xrightarrow{\text{a.s.}} \Pi_T^{(\infty)} \quad \text{and} \quad \mathbb{P}(\Pi_T^{(\infty)} < 0) = 1.$$

Definition 3 (Convergence in probability to loss). *A sequence $(\Pi_T^{(n)})_{n \geq 1}$ converges in probability to loss if there exists a random variable $\Pi_T^{(\infty)}$ such that*

$$\Pi_T^{(n)} \xrightarrow{\mathbb{P}} \Pi_T^{(\infty)} \quad \text{and} \quad \mathbb{P}(\Pi_T^{(\infty)} < 0) = 1.$$

Remark 1. *These notions satisfy the hierarchy*

$$\text{almost-sure loss at } T \implies$$

$$\text{almost-sure convergence to loss} \implies$$

$$\text{convergence in probability to loss}.$$

For the assignment, only the first notion answers the question literally.

1.3 Decomposition of net profit and loss

For any strategy in a market with frictions, write

$$\Pi_T = \Pi_T^{\text{gross}} - \Phi_T,$$

where Π_T^{gross} is the gross payoff of the position and $\Phi_T \geq 0$ is the cumulative frictional cost. The latter may include transaction costs, commissions, financing spreads, borrow/management fees, or market impact. This separates two distinct mechanisms for loss:

- (i) *directional or contractual loss*: the gross payoff is itself negative;
- (ii) *frictional loss*: the gross payoff is neutral or small, but costs dominate.

This distinction matters because a negative friction term alone is usually not enough to prove almost-sure loss unless the gross payoff has first been neutralized.

2 Part 1: Candidate strategies

We now examine several candidate strategies. Some satisfy the assignment literally; others only support weaker probabilistic claims.

2.1 Strategy I: Negative carry between borrowing and lending

Construction. Fix $\tau = 9/12$. At time 0, borrow $W > 0$ at a fixed annualized rate r_B and lend the proceeds at a fixed annualized rate r_L , with

$$r_L < r_B.$$

Hold both positions to maturity.

Terminal payoff. Under simple accrual,

$$\Pi = W(1 + r_L\tau) - W(1 + r_B\tau) = W(r_L - r_B)\tau.$$

Claim 1. *If $r_L < r_B$, then $\Pi < 0$ almost everywhere.*

Proof. Since $W > 0$, $\tau > 0$, and $r_L - r_B < 0$,

$$\Pi = W(r_L - r_B)\tau < 0.$$

Payoff is fully determined at inception, hence a.s. $\square$

Interpretation. Once the negative spread is locked in contractually, nothing stochastic remains.

2.2 Strategy II: Buying a zero-coupon bond above redemption value

Construction. Let a default-free zero-coupon bond mature within the horizon and redeem at deterministic

value $F > 0$. Suppose it is purchased at time 0 for price $P_0 > F$ and held to maturity. $\square$

Terminal payoff.

$$\Pi = F - P_0.$$

Claim 2. *If $P_0 > F$, then the position is an almost-sure loser. In fact, Π is a negative constant.*

Proof. The terminal payoff is exactly F , so

$$\Pi = F - P_0 < 0.$$

Again, there is no residual uncertainty. $\square$

Interpretation. This is mathematically valid, though somewhat artificial economically unless one explicitly allows off-market execution or embedded fees.

2.3 Strategy III: Fee-dominated risk-free investment

Construction. Invest wealth $W > 0$ in a risk-free asset with deterministic return factor $1 + R_f$ over the horizon, and pay a deterministic proportional fee $\phi > R_f$.

Terminal payoff.

$$\Pi = W(R_f - \phi).$$

Claim 3. *If $\phi > R_f$, then the strategy is an almost-sure loser. In fact, Π is a negative constant.*

Proof. Since $W > 0$ and $\phi > R_f$,

$$\Pi = W(R_f - \phi) < 0.$$

The return and fee are both contractual, so the payoff is deterministic. $\square$

Interpretation. This is the same logic as Strategy I: a deterministic negative spread survives after all uncertainty has been removed.

2.4 Strategy IV: Perfect hedge with deterministic borrow cost

Construction. Construct a portfolio whose gross terminal payoff is exactly 0, but requires paying a deterministic borrow fee $Wb\tau$ with $b > 0$ over horizon.

Terminal payoff.

$$\Pi = -Wb\tau.$$

Claim 4. *If the hedge eliminates all price risk and the borrow cost is deterministic with $b > 0$, then the strategy is an almost-sure loser.*

Proof. By assumption, the gross payoff is identically zero, so only the borrow fee remains:

$$\Pi = -Wb\tau < 0.$$

Interpretation. The point is not the short position itself, but the fact that all upside and downside have been hedged away, leaving only a deterministic carrying cost.

Remark 2. *If the hedge is not exact, the conclusion weakens immediately: the gross payoff becomes state-dependent, and almost-sure loss no longer follows automatically.*

2.5 Strategy V: Repeated high-turnover trading with transaction costs

Construction. At dates $t_0 < t_1 < \dots < t_N$, repeatedly enter and exit positions, paying deterministic cost $c_k > 0$ in round k . Let G_k denote the gross payoff before cost. Then

$$\Pi = \sum_{k=1}^N G_k - \sum_{k=1}^N c_k.$$

Claim 5. *Positive transaction costs alone do not imply almost-sure loss unless the gross payoffs are already controlled strongly enough to be dominated by costs in every state.*

Proof. The cost term is always negative, but the gross term remains stochastic in general. Without an almost-sure upper bound

$$\sum_{k=1}^N G_k < \sum_{k=1}^N c_k,$$

one cannot rule out states in which $\Pi > 0$. $\square$

Interpretation. Transaction costs make a strategy worse, but they do not by themselves make it a guaranteed loser.

High-turnover trading is not, by itself, a literal answer to the assignment.

2.6 Strategy VI: Large trading with adverse market impact

Construction. Suppose each trade incurs positive impact cost $I_k > 0$, while G_k denotes the gross trading profit before impact. Then

$$\Pi = \sum_{k=1}^N G_k - \sum_{k=1}^N I_k.$$

Claim 6. *Adverse market impact alone does not imply almost-sure loss.*

Proof. Exactly as above, the impact term is negative but the gross payoff remains state-dependent. Unless the gains are already neutralized or dominated almost surely, some states may still yield $\Pi > 0$. $\square$

2.7 Strategy VII: Leveraged exposure and volatility drag

Construction. Consider a dynamically rebalanced leveraged position in a risky asset. Continuous-time approximations suggest long-run log-growth as

$$L\mu - \frac{1}{2}L^2\sigma^2.$$

Claim 7. *Negative expected log-growth does not imply almost-sure loss at a fixed finite horizon.*

Proof. A negative growth-rate calculation is an expectation statement. It does not rule out sample paths on which the realized terminal payoff is positive. Therefore it does not establish $\mathbb{P}(\Pi_T < 0) = 1$. $\square$

2.8 Strategy VIII: Long options with $\theta < 0$

Construction. Take a long option position with negative theta.

Claim 8. *Negative theta alone does not imply almost-sure loss over the horizon.*

Proof. Theta is a local sensitivity with respect to time, holding other state variables fixed. Realized option P&L depends on the full underlying path, and favorable price moves may over offset time decay. $\square$

2.9 Strategy IX: Gambler's ruin w. negative edge

Construction. Suppose wealth evolves through repeated bets with negative expected return.

Claim 9. *Eventual ruin results do not imply almost-sure loss at a fixed nine-month horizon.*

Proof. Gambler's ruin is an asymptotic or infinite-horizon statement. Over a fixed finite horizon, favorable paths remain possible, so one cannot conclude $\mathbb{P}(\Pi_T < 0) = 1$ from eventual ruin alone. $\square$

3 Synthesis of Part 1

The strategies that satisfy the assignment are those in which all stochastic exposure has been removed and a strictly negative contractual spread remains.

By contrast, transaction costs, market impact, leverage, theta decay, and gambler's ruin do not by themselves imply almost-sure loss at the fixed horizon.

4 Part 2: Does flipping the sign produce a guaranteed winner?

Write the original payoff as

$$\Pi = \Pi^{\text{gross}} - \Phi.$$

If every position is reversed, the gross term changes sign while the friction term does not:

$$\Pi^{\text{flip}} = -\Pi^{\text{gross}} - \Phi.$$

So, in general, reversing the sign does *not* turn a loser into a winner. If the original strategy loses because of sign-invariant frictions, those remain after reversal.

However, if the original loss comes from a deterministic directional spread, then sign reversal does produce a deterministic winner.

4.1 Example: reversing Strategy I

For Strategy I,

$$\Pi = W(r_L - r_B)\tau < 0.$$

Reversing the legs gives

$$\Pi^{\text{flip}} = W(r_B - r_L)\tau > 0.$$

The sign-reversal of a deterministic negative-carry strategy is a deterministic winner.

4.2 General conclusion for Part 2

Flipping the sign of a losing strategy produces a guaranteed winner only when the original loss comes from a deterministic directional spread. It does not do so when the loss comes from sign-invariant frictions.

Proof. From

$$\Pi = \Pi^{\text{gross}} - \Phi \quad \implies \quad \Pi^{\text{flip}} = -\Pi^{\text{gross}} - \Phi,$$

the term remains unchanged. Therefore reversal helps only when the original negative payoff comes from the sign of the gross component rather than from the cost term. $\square$

5 Conclusion

The clean answer to the assignment is to separate true guaranteed loss from weaker probabilistic notions. Many familiar mechanisms—transaction costs, market impact, leverage, theta decay, gambler's ruin—can make a strategy unattractive, and sometimes asymptotically loss-making, but they don't by themselves imply guaranteed loss over a fixed nine-month horizon.

The valid examples are the ones with no residual market uncertainty: once uncertainty is hedged away, a strictly negative contractual spread remains. In that class, negative carry, overpriced purchase of a deterministic cash flow, fee-dominated risk-free investment, and perfectly hedged positions with deterministic carrying costs all work.

Likewise, flipping the sign does not usually turn a loser into a winner. It works only when the original loss comes from a directional spread, not when it comes from frictions that survive reversal.